

Problem Statement

01 Lecturers Lack Time to Prepare Interactive Learning Materials

University lecturers face heavy workloads that limit the time available to prepare interactive learning activities. According to The University and College Union (2016), academic staff work an average of 50.9 hours per week, with 28.5% averaging more than 55 hours per week, leaving limited time for lesson preparation.

02 Students Experience Passive Learning During Lectures

Traditional lecture-based teaching often results in passive student participation, making it difficult for lecturers to assess students' understanding during class. According to Freeman et al. (2014), active learning significantly improves academic performance and reduces failure rates compared to traditional lectures.

03 Existing AI Learning Systems Do Not Support Live Classroom Adaptation

Although existing AI learning systems can generate learning materials and quizzes, they mainly focus on content preparation and lack integrated support for real-time classroom interaction, slide synchronisation, anonymous questioning and adaptive learning support (Zeghouani et al., 2024; Xie et al., 2026). This highlights the need for a unified platform that supports both pre-class preparation and live classroom activities.